Week 11 - Graded Assignment 11

The due date for submitting this assignment has passed.
Due on 2026-04-29, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-29, 11:51 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

1 point
Consider the following function

$$f(x,y,z) = x^2 z + y^3 e^{xy} + \sin(xyz).$$

Which of the following option(s) is (are) true?

- ( ) $f_{xx} = 2z + y^5 e^{xy} - yz\sin(xyz)$
- (•) $f_{xx} = 2z + y^5 e^{xy} - (yz)^2\sin(xyz)$
- ( ) $f_{zy} = \cos(xyz)x^2 yz$
- ( ) $f_{xz} = 2x - xy^2 z\sin(xyz)$

Yes, the answer is correct.
Score: 1
Accepted Answers:
$f_{xx} = 2z + y^5 e^{xy} - (yz)^2\sin(xyz)$

Number of critical points of the function $f(x,y,z) = e^x(x^2 - y^2 - 2z^2)$ is

2

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 2

1 point
1 point
Choose the correct statement(s) about $f(x,y) = \sqrt{x^2 + y^2}$.

- [ ] f has no critical points.
- [x] $(0,0)$ is the only critical point of f.
- [ ] $(0,0)$ is a local maximum of f.
- [x] $(0,0)$ is a local minimum of f.
- [x]

$(0,0)$ is a global minimum of f.

Yes, the answer is correct.
Score: 1
Accepted Answers:

$(0,0)$ is the only critical point of f.

$(0,0)$ is a local minimum of f.

$(0,0)$ is a global minimum of f.

Find the minimum value of the surface area of a rectangular box that has a volume of 1000 unit^3.

$Note$: If x, y and z represent the lengths of the three sides of a rectangular box, then the surface area and volume are given by $2(xy + yz + zx)$ and xyz, respectively.

600

Yes, the answer is correct.
Score: 1
Accepted Answers:

(Type: Numeric) 600

1 point

1 point

Consider a function $f(x, y, z) = x^3 + 3y^3 + z^3 - 4x - 6y - 4z$. If A is the the Hessian matrix of $f(x, y, z)$ at (0,1,1), then which of the following option(s) is(are) true?

- [x] $det(A) = 0$
- [x] $Rank(A) = 2$
- [x] $Nullity(A) = 1$
- [ ] $det(A) = 648$
- [ ] $Rank(A) = 3$

Yes, the answer is correct.
Score: 1
Accepted Answers:

$det(A) = 0$

$Rank(A) = 2$

$Nullity(A) = 1$

1 point

Consider the following function

$$f(x, y) = e^x + e^y - e^{x+y}$$

Which of the following option(s) is (are) true?

- [ ] Hessian matrix of f at point $(1,1)$ is $\begin{bmatrix} 1-e & -e^2 \\ -e^2 & 1-e \end{bmatrix}$.
- [x] Hessian matrix of f at point $(1,1)$ is $\begin{bmatrix} e-e^2 & -e^2 \\ -e^2 & e-e^2 \end{bmatrix}$.
- [ ]

f has a point of maximum.

- [x] f has a saddle point.

Yes, the answer is correct.
Score: 1
Accepted Answers:
Hessian matrix of f at point $(1, 1)$ is $\begin{bmatrix} e - e^2 & -e^2 \\ -e^2 & e - e^2 \end{bmatrix}$.

f has a saddle point.

1 point
Choose the correct statements.

- [ ] If $f: R^3 \to R$ then the Hessian matrix $H_f(x, y, z)$ is same for all $(x, y, z) \in R^3$.
- [x] If $f: R^3 \to R$ is a linear transformation then the Hessian matrix $H_f(x, y, z)$ is same for all $(x, y, z) \in R^3$.
- [x] For each $n \in N$, there exists a map $f: R^n \to R$ such that the rank of the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$ is n for some $(x_1, x_2, \ldots, x_n) \in R^n$.
- [x] For each $n \in N$, there exists a map $f: R^n \to R$ such that the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$ is symmetric for all $(x_1, x_2, \ldots, x_n) \in R^n$.

Yes, the answer is correct.
Score: 1
Accepted Answers:
If $f: R^3 \to R$ is a linear transformation then the Hessian matrix $H_f(x, y, z)$ is same for all $(x, y, z) \in R^3$.
For each $n \in N$, there exists a map $f: R^n \to R$ such that the rank of the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$ is n for some $(x_1, x_2, \ldots, x_n) \in R^n$.
For each $n \in N$, there exists a map $f: R^n \to R$ such that the Hessian matrix $H_f(x_1, x_2, \ldots, x_n)$ is symmetric for all $(x_1, x_2, \ldots, x_n) \in R^n$.

The maximum value of the function $-2x^2 + 2xy - y^2 + 12x - 4y - 7$ is
13
Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 13
1 point
1 point
Define a function

$$f(x, y) = \begin{cases} (x - y)^2 \sin\left(\frac{1}{x - y}\right) & \text{if } x \neq y \\ 0 & \text{if } x = y. \end{cases}$$

Choose the set of correct options.

- [x] f is continuous at $(0,0)$.
- [ ] $f_x(0,0) \neq f_y(0,0)$
- [x] $f_x(0,0) = f_y(0,0)$
- [x] f is differentiable at $(0,0)$.

Yes, the answer is correct.
Score: 1
Accepted Answers:
f is continuous at $(0,0)$.
$f_x(0,0) = f_y(0,0)$
f is differentiable at $(0,0)$.

A company produces two types of chocolates, one that sells for Rs.3 and another at Rs.2. The total revenue, in lakhs, from the sale of x lakhs chocolates at Rs.2 each and y lakhs at Rs.3 each is given by

$$F(x,y) = 2x + 3y.$$

The total cost in lakhs for producing x lakhs of the Rs.2 chocolates and y lakhs of the Rs.3 chocolates is given by

$$G(x,y) = 2x^2 - 2xy + y^2 - 9x + 6y + 7.$$

Then the maximum profit in lakhs is

[Enter your answer upto two decimal places]
11.25

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 11.25
1 point

1 point
(MSQ) Let $f: \mathbb{R}^2 \to \mathbb{R}$ be a multivariable function and $(x_0, y_0) \in \mathbb{R}^2$.
Consider the following statements:

- P: f is continuous at (x_0, y_0).
- Q: The partial derivatives of f exist and are continuous in a neighborhood of (x_0, y_0).
- R: The tangent plane to the graph of f exists at (x_0, y_0).

Choose the correct statements from the following.

- [ ] P implies that f is differentiable at (x_0, y_0).
- [x] If f is differentiable at (x_0, y_0), then P is satisfied.
- [x] Q implies that f is differentiable at (x_0, y_0).

- [ ] If f is differentiable at (x_0, y_0), then Q is satisfied.
- [x] R implies that f is differentiable at (x_0, y_0).
- [x] If f is differentiable at (x_0, y_0), then R is satisfied.

Yes, the answer is correct.
Score: 1
Accepted Answers:
If f is differentiable at (x_0, y_0), then P is satisfied.
Q implies that f is differentiable at (x_0, y_0).
R implies that f is differentiable at (x_0, y_0).
If f is differentiable at (x_0, y_0), then R is satisfied.

(NAT) A particle moving in a potential field tends to settle at points where the potential energy is a local minimum (stable equilibrium point). Consider a particle moving in R^2 where the potential energy as a function of position is given by

$$V(x, y) = x^4 + y^4 - 4xy + 1,$$

for all $(x, y) \in R^2$. Using the Hessian test, find the potential energy at stable equilibrium points.
-1
Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) -1
1 point